CASE STUDY

Cybersecurity Incident Report

Network Traffic Analysis

PART 1

Provide a summary of the problem found in the DNS and ICMP traffic log.

Several users reported that they could not access www.yummyrecipesforme.com and received a "destination port unreachable" error. I reproduced the problem and captured the traffic with tcpdump.

The capture showed DNS queries leaving the client through UDP toward the DNS server. Instead of a normal DNS response, the client received ICMP messages reporting "udp port 53 unreachable."

Because port 53 is used by DNS, the traffic shows that the DNS service could not be reached. ICMP was reporting the failure rather than causing it.

KEY EVIDENCE

UDP request to DNS | ICMP response | "udp port 53 unreachable"

Affected service: DNS

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PART 2

Explain your analysis of the data and provide at least one possible cause of the incident.

TIME INCIDENT OCCURRED

The traffic capture shows the incident at approximately 1:24 p.m.

HOW THE TEAM BECAME AWARE

Customers reported that the website would not load and displayed a destination port unreachable error. The same problem was reproduced during the investigation.

ACTIONS TAKEN TO INVESTIGATE

I opened tcpdump and tried to load the website again so I could observe the packets sent and received during the failure.

KEY FINDINGS

The client sent DNS queries through UDP to the DNS server. The replies were ICMP error messages stating that UDP port 53 was unreachable. The same result appeared more than once.

LIKELY CAUSE

The immediate issue is that the DNS service was unavailable or could not be reached on port 53. A service failure or a firewall configuration could cause this. A DoS attack is also possible, but this capture alone does not prove that one occurred.

NEXT CHECK

I would verify that the DNS service is running and then check the firewall rules for port 53. If those look normal, I would review server and network logs for signs of abnormal traffic.